

## Practical 8a.

```
niyathi@LAPTOP-NIYATHI:~/OSLab$ gcc practical8a.c -o practical8a
niyathi@LAPTOP-NIYATHI:~/OSLab$ ./practical8a
Enter source file name:
practical1.c
Enter destination file for low-level copy: fork.c
Enter destination file for standard copy: fork1.c

Low-level copy completed.
Time taken: 0.000595 seconds

Standard library copy completed.
Time taken: 0.000079 seconds

--- PERFORMANCE COMPARISON ---
Standard library I/O was faster.
```

## Practical 8b.

```
niyathi@LAPTOP-NIYATHI:~/OSLab$ gcc -Wall -g practical8b.c -o practical8b
niyathi@LAPTOP-NIYATHI:~/OSLab$ ./practical8b
=====
COPY-ON-WRITE DEMONSTRATION
=====

Parent PID: 5135

Memory initialized in parent.

[PARENT - BEFORE fork() - PID 5135]
VmSize:    13008 kB
VmRSS:     12188 kB

Creating child process using fork()...

=====
PARENT PROCESS
=====
Parent PID: 5135
Child PID : 5136

=====
CHILD PROCESS
=====
Child PID: 5136

[CHILD - AFTER fork(), BEFORE modification - PID 5136]
VmSize:    13008 kB
```

```
=====
Child PID: 5136

[CHILD - AFTER fork(), BEFORE modification - PID 5136]
VmSize:    13008 kB
VmRSS:     11244 kB

Child is modifying the shared memory...
Child modification completed.

[CHILD - AFTER modification - PID 5136]
VmSize:    13008 kB
VmRSS:     11244 kB

Child data[0] = B
Child exiting...

Child has completed.

[PARENT - AFTER child modification - PID 5135]
VmSize:    13008 kB
VmRSS:     12188 kB

Parent data[0] = A
Child exit status: 0

=====
COPY-ON-WRITE TEST COMPLETED
=====
```